Lab 2 - Greedy regret heuristics

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Description of the problem

Each instance contains nodes with coordinates (x, y) and a cost. Select exactly half of the nodes (rounded up) and connect them into a closed cycle. Minimise **cycle length + sum of the costs of the selected nodes**. Distances are Euclidean, rounded to the nearest integer.

Algorithms

200 solutions per method (one starting from each node). Both methods insert at any position (path or cycle edge), but pick the node by score instead of the smallest delta.

For an unvisited node u: `delta(u, p)` = added length + cost(u) at position p, `best` = smallest delta, `regret` = second smallest delta - best.
Score: **2-regret** = regret; **2-regret + best change** = regret - best. The node with the highest score is inserted at its best position.

```
Nearest neighbour, any position, 2-regret (start):
  tour = [start]
  while |tour| < k:
      for each unvisited u: compute best and regret over all path positions
      insert the u with the highest score at its best position

Greedy cycle, 2-regret (start):
  v = unvisited node minimising dist(start, v) + cost(v);  tour = [start, v]
  while |tour| < k:
      for each unvisited u: compute best and regret over all cycle edges
      insert the u with the highest score at its best edge
```

Results

Objective = cycle length + sum of node costs, over 200 runs per method. The best value in each column is in bold.
All best solutions were checked with the solution checker (the same computation as the lecturer's Solution checker.xlsx): **yes**.
Results are identical in all implemented languages: **yes**.

Instance TSPA

Method	Min	Max	Avg	Best solution checked
Random solution	237131	287973	263514	yes
Nearest neighbor (end only)	84362	90075	87477	yes
Nearest neighbor (any position)	72328	77022	74347	yes
Greedy cycle	72590	**75281**	73797	yes
Nearest neighbor (any position), 2-regret	111527	133181	122861	yes
Nearest neighbor (any position), 2-regret + best change	72419	78504	75561	yes
Greedy cycle, 2-regret	112654	130924	121007	yes
Greedy cycle, 2-regret + best change	**72011**	75532	**73350**	yes

Instance TSPB

Method	Min	Max	Avg	Best solution checked
Random solution	188214	231879	206802	yes
Nearest neighbor (end only)	56007	63166	59397	yes
Nearest neighbor (any position)	52915	66164	59475	yes
Greedy cycle	53016	63736	59709	yes
Nearest neighbor (any position), 2-regret	69032	80767	74763	yes
Nearest neighbor (any position), 2-regret + best change	**50485**	**61406**	**55698**	yes
Greedy cycle, 2-regret	69455	78142	73403	yes
Greedy cycle, 2-regret + best change	52906	63326	57751	yes

Running time

Average time of one run of the algorithm only, over the same runs as above. The fastest language in each row is in bold.

Instance TSPA

Method	Python	C	C++	Rust
Random solution	77.6 µs	380 ns	321 ns	**295 ns**
Nearest neighbor (end only)	374.0 µs	**14.0 µs**	20.4 µs	23.5 µs
Nearest neighbor (any position)	24.39 ms	534.2 µs	**481.1 µs**	567.2 µs
Greedy cycle	22.58 ms	687.4 µs	**462.6 µs**	648.7 µs
Nearest neighbor (any position), 2-regret	42.75 ms	**688.7 µs**	2.96 ms	3.21 ms
Nearest neighbor (any position), 2-regret + best change	44.34 ms	**678.8 µs**	2.86 ms	3.14 ms
Greedy cycle, 2-regret	43.19 ms	**942.5 µs**	2.80 ms	1.41 ms
Greedy cycle, 2-regret + best change	43.23 ms	**1.08 ms**	2.78 ms	1.39 ms

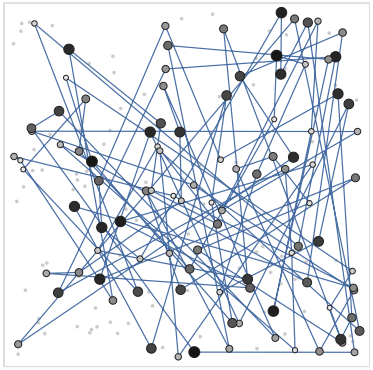
Instance TSPB

Method	Python	C	C++	Rust
Random solution	70.2 µs	**425 ns**	481 ns	500 ns
Nearest neighbor (end only)	360.0 µs	**15.6 µs**	24.0 µs	18.3 µs
Nearest neighbor (any position)	23.51 ms	529.4 µs	**515.9 µs**	561.6 µs
Greedy cycle	21.92 ms	541.8 µs	**464.2 µs**	639.2 µs
Nearest neighbor (any position), 2-regret	44.36 ms	**679.2 µs**	2.78 ms	3.66 ms
Nearest neighbor (any position), 2-regret + best change	43.22 ms	**704.9 µs**	2.91 ms	3.14 ms
Greedy cycle, 2-regret	43.15 ms	**941.7 µs**	2.80 ms	1.48 ms
Greedy cycle, 2-regret + best change	42.78 ms	**941.3 µs**	2.76 ms	1.40 ms

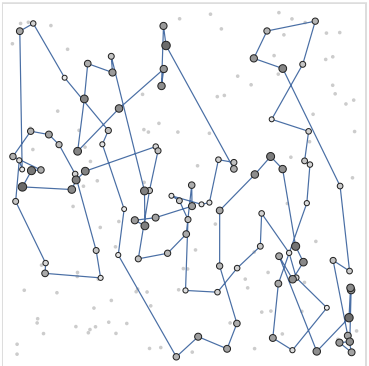
Visualisation of the best solutions

Node colour and size show the node cost (dark and large = expensive). Grey dots are nodes not selected.

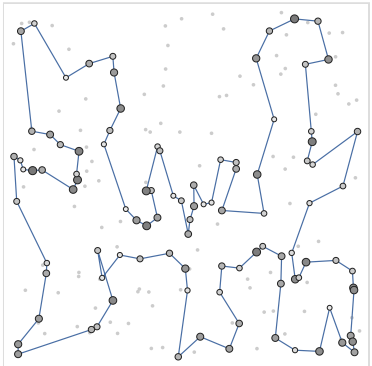
Instance TSPA



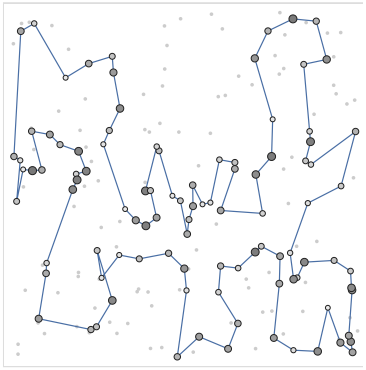
Random solution
237131 (length 140636 + cost 96495)



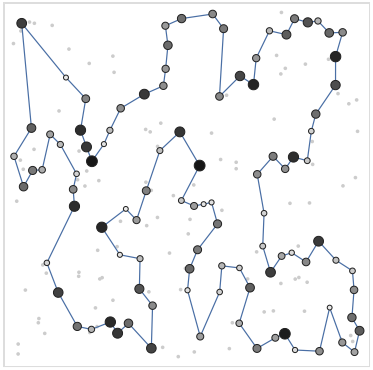
Nearest neighbor (end only)
84362 (length 31319 + cost 53043)



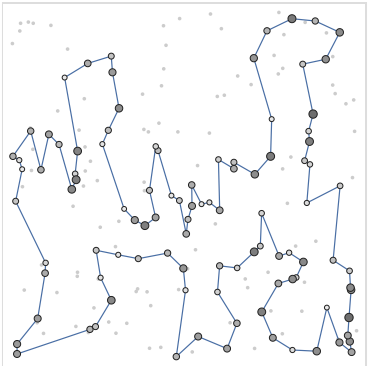
Nearest neighbor (any position)
72328 (length 21816 + cost 50512)



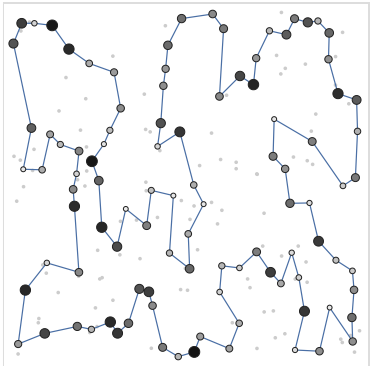
Greedy cycle
72590 (length 21570 + cost 51020)



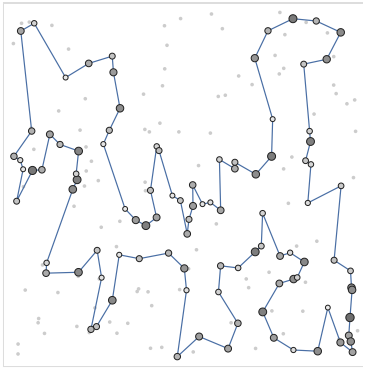
Nearest neighbor (any position), 2-regret
111527 (length 19974 + cost 91553)



Nearest neighbor (any position), 2-regret + best change
72419 (length 20127 + cost 52292)

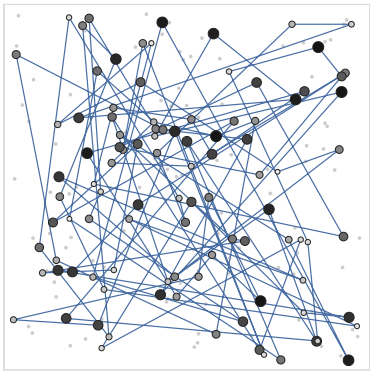


Greedy cycle, 2-regret
112654 (length 21632 + cost 91022)

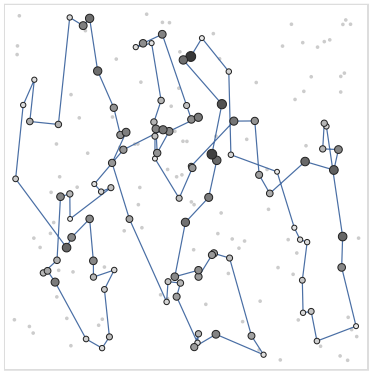


Greedy cycle, 2-regret + best change
72011 (length 20248 + cost 51763)

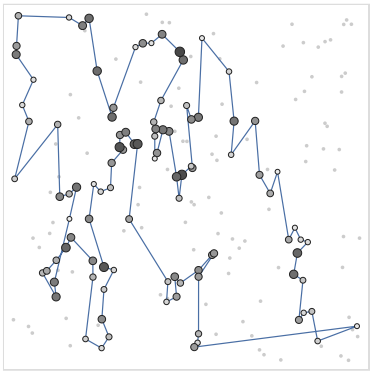
Instance TSPB



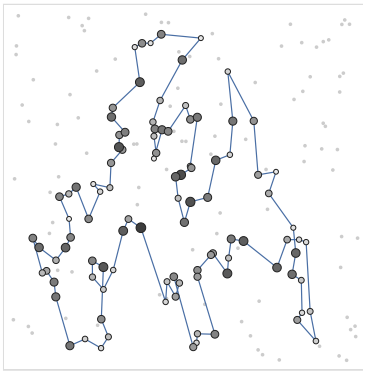
Random solution
188214 (length 138112 + cost 50102)



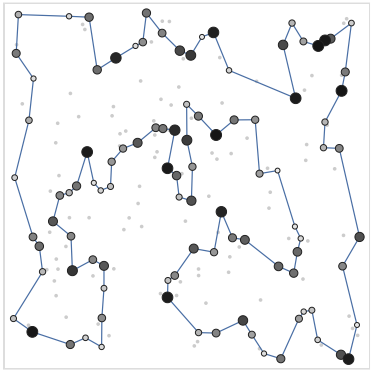
Nearest neighbor (end only)
56007 (length 24632 + cost 31375)



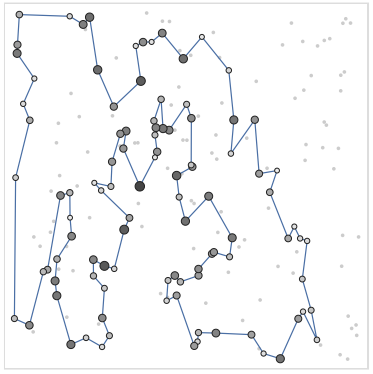
Nearest neighbor (any position)
52915 (length 20068 + cost 32847)



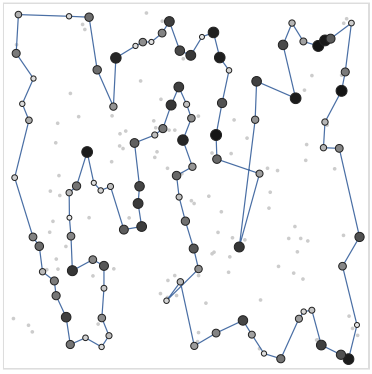
Greedy cycle
53016 (length 16704 + cost 36312)



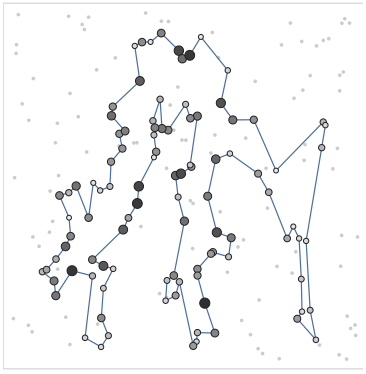
Nearest neighbor (any position), 2-regret
69032 (length 21267 + cost 47765)



Nearest neighbor (any position), 2-regret + best change
50485 (length 18835 + cost 31650)



Greedy cycle, 2-regret
69455 (length 20736 + cost 48719)



Greedy cycle, 2-regret + best change
52906 (length 16675 + cost 36231)

Best solutions (0-based node indices)

Instance TSPA

Random solution
128, 34, 55, 103, 152, 57, 106, 85, 120, 112, 86, 43, 48, 96, 63, 38, 100, 61, 164, 195, 178, 194, 58, 117, 6, 158, 173, 40, 37, 121, 156, 192, 5, 101, 165, 1, 29, 131, 147, 60, 52, 170, 64, 54, 142, 93, 41, 88, 107, 159, 80, 68, 20, 36, 15, 22, 12, 146, 67, 116, 149, 189, 137, 133, 35, 99, 51, 30, 13, 154, 122, 75, 155, 33, 115, 162, 23, 87, 197, 56, 19, 78, 118, 82, 66, 182, 139, 190, 144, 161, 127, 65, 191, 50, 114, 45, 16, 26, 163, 129

Nearest neighbor (end only)
54, 9, 33, 36, 177, 67, 144, 53, 135, 58, 134, 176, 174, 106, 89, 107, 57, 182, 151, 93, 50, 3, 1, 164, 178, 29, 136, 186, 184, 47, 159, 130, 153, 78, 88, 193, 170, 146, 75, 64, 150, 42, 113, 43, 85, 142, 8, 131, 84, 158, 192, 181, 114, 141, 180, 132, 76, 80, 48, 31, 63, 149, 168, 97, 127, 187, 30, 123, 152, 108, 60, 116, 49, 11, 119, 160, 83, 90, 51, 16, 35, 87, 45, 140, 52, 179, 185, 28, 102, 156, 157, 71, 91, 38, 100, 4, 112, 139, 115, 5

Nearest neighbor (any position)
104, 129, 69, 127, 97, 168, 116, 149, 63, 11, 49, 160, 83, 123, 100, 152, 108, 60, 31, 48, 80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 156, 51, 16, 67, 144, 140, 53, 179, 52, 177, 36, 33, 9, 54, 131, 174, 8, 164, 186, 136, 163, 29, 101, 178, 1, 169, 3, 50, 147, 47, 93, 106, 42, 89, 113, 130, 159, 146, 170, 153, 193, 78, 88, 107, 64, 57, 182, 151, 150, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45, 87, 141, 187, 30, 189, 2, 171

Greedy cycle
31, 116, 149, 168, 63, 11, 49, 60, 108, 152, 100, 90, 123, 83, 160, 97, 127, 69, 171, 2, 189, 30, 187, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 151, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 113, 89, 42, 106, 93, 47, 147, 50, 3, 169, 1, 178, 101, 29, 163, 136, 186, 164, 142, 8, 174, 131, 54, 9, 33, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 156, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48

Nearest neighbor (any position), 2-regret
199, 82, 116, 119, 11, 49, 108, 152, 123, 160, 12, 97, 120, 92, 171, 166, 183, 10, 22, 98, 27, 87, 141, 23, 180, 28, 156, 16, 190, 74, 144, 179, 177, 36, 99, 41, 61, 135, 181, 58, 84, 134, 65, 158, 111, 182, 105, 57, 64, 107, 88, 78, 26, 75, 170, 159, 130, 117, 113, 106, 150, 0, 176, 174, 8, 142, 85, 128, 3, 1, 59, 46, 86, 55, 17, 29, 19, 163, 24, 136, 186, 109, 34, 62, 154, 39, 175, 115, 5, 112, 139, 18, 4, 76, 132, 68, 44, 14, 38, 80

Nearest neighbor (any position), 2-regret + best change
171, 2, 189, 187, 30, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 174, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 75, 170, 146, 159, 130, 47, 93, 50, 3, 169, 1, 59, 178, 101, 55, 29, 163, 136, 186, 164, 142, 8, 9, 54, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 100, 123, 83, 160, 152, 108, 49, 60, 116, 149, 63, 168, 97, 127, 69, 129, 104

Greedy cycle, 2-regret
190, 52, 177, 140, 61, 45, 67, 35, 102, 180, 145, 23, 162, 68, 132, 76, 4, 71, 91, 137, 126, 48, 199, 77, 82, 63, 49, 108, 152, 100, 123, 160, 12, 161, 97, 188, 129, 110, 92, 171, 166, 183, 10, 27, 70, 98, 25, 114, 118, 181, 192, 158, 58, 84, 134, 72, 0, 151, 106, 89, 124, 57, 64, 107, 193, 75, 170, 159, 130, 117, 93, 81, 85, 142, 164, 169, 47, 37, 147, 21, 155, 101, 17, 29, 19, 163, 24, 136, 186, 109, 34, 62, 154, 39, 175, 5, 112, 139, 133, 51

Greedy cycle, 2-regret + best change
80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 51, 16, 67, 144, 140, 53, 179, 52, 177, 36, 131, 33, 54, 9, 8, 142, 164, 186, 136, 163, 29, 55, 101, 178, 1, 169, 3, 50, 93, 47, 130, 159, 146, 170, 75, 153, 193, 78, 88, 107, 64, 57, 182, 138, 151, 42, 89, 113, 106, 150, 174, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45, 87, 141, 189, 2, 171, 187, 30, 161, 127, 97, 160, 83, 123, 100, 152, 108, 49, 11, 168, 63, 149, 116, 60, 31, 48

Instance TSPB

Random solution

80, 23, 16, 71, 163, 97, 140, 42, 136, 144, 107, 193, 172, 44, 12, 15, 105, 190, 160, 98, 151, 116, 103, 45, 147, 195, 177, 60, 110, 64, 153, 55, 73, 106, 188, 137, 17, 170, 50, 8, 13, 6, 88, 104, 0, 133, 38, 184, 61, 4, 113, 5, 68, 20, 111, 134, 129, 39, 191, 59, 148, 54, 63, 158, 96, 21, 198, 171, 95, 142, 1, 181, 27, 186, 34, 51, 161, 18, 75, 46, 130, 154, 56, 77, 176, 162, 139, 199, 57, 90, 2, 91, 185, 19, 168, 40, 132, 84, 66, 30

Nearest neighbor (end only)

32, 89, 57, 76, 54, 127, 129, 90, 153, 20, 14, 2, 110, 118, 56, 114, 196, 10, 148, 143, 78, 192, 162, 104, 185, 184, 3, 151, 69, 172, 133, 120, 96, 121, 198, 48, 72, 46, 86, 117, 160, 136, 8, 21, 154, 140, 47, 170, 125, 177, 6, 132, 29, 197, 175, 134, 45, 60, 40, 71, 155, 97, 115, 180, 65, 28, 149, 145, 158, 116, 147, 42, 50, 130, 30, 183, 38, 64, 85, 59, 167, 51, 87, 179, 82, 88, 171, 99, 67, 24, 34, 181, 152, 73, 98, 27, 135, 75, 79, 156

Nearest neighbor (any position)

110, 2, 14, 20, 74, 153, 182, 52, 90, 129, 127, 80, 54, 192, 162, 104, 76, 185, 57, 89, 86, 46, 72, 3, 184, 173, 151, 122, 117, 8, 160, 69, 172, 136, 133, 120, 156, 43, 48, 96, 198, 121, 130, 191, 50, 42, 147, 116, 37, 146, 13, 149, 28, 145, 65, 158, 177, 125, 26, 170, 197, 36, 132, 29, 6, 180, 115, 155, 71, 175, 134, 45, 157, 60, 40, 141, 97, 154, 140, 47, 38, 21, 176, 30, 183, 189, 168, 64, 59, 85, 51, 73, 167, 152, 34, 24, 181, 179, 87, 82

Greedy cycle

49, 64, 33, 40, 60, 71, 155, 141, 157, 45, 134, 175, 62, 36, 197, 29, 132, 178, 103, 6, 180, 115, 170, 177, 125, 26, 97, 140, 154, 47, 38, 21, 176, 30, 183, 191, 130, 63, 121, 198, 96, 48, 89, 156, 120, 133, 136, 172, 160, 69, 8, 117, 72, 46, 86, 184, 3, 173, 122, 151, 98, 111, 27, 135, 76, 185, 57, 104, 162, 54, 192, 127, 52, 182, 153, 20, 74, 2, 14, 90, 129, 80, 150, 84, 66, 67, 31, 24, 34, 152, 181, 88, 179, 87, 82, 73, 167, 51, 59, 85

Nearest neighbor (any position), 2-regret

57, 44, 89, 32, 43, 48, 58, 198, 121, 195, 50, 42, 116, 37, 13, 149, 145, 65, 178, 103, 132, 1, 108, 62, 175, 134, 157, 60, 141, 155, 113, 115, 19, 177, 125, 26, 161, 154, 140, 47, 38, 21, 168, 136, 8, 95, 128, 122, 151, 111, 3, 137, 72, 86, 0, 185, 104, 162, 54, 127, 129, 52, 182, 150, 84, 66, 35, 24, 25, 73, 59, 101, 179, 88, 163, 99, 171, 17, 74, 20, 14, 2, 93, 23, 110, 118, 131, 148, 10, 196, 193, 61, 105, 169, 165, 199, 70, 190, 194, 12

Nearest neighbor (any position), 2-regret + best change

65, 145, 28, 149, 13, 146, 37, 116, 147, 42, 50, 130, 63, 121, 198, 96, 48, 156, 89, 57, 185, 76, 104, 162, 54, 192, 80, 127, 129, 90, 153, 14, 2, 20, 74, 17, 171, 99, 88, 179, 87, 82, 51, 85, 59, 73, 167, 181, 152, 34, 24, 67, 66, 27, 98, 151, 122, 3, 184, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 21, 183, 30, 38, 47, 154, 140, 64, 33, 40, 141, 155, 71, 60, 157, 45, 134, 175, 62, 36, 197, 6, 115, 170, 125, 177, 29, 132, 174, 1

Greedy cycle, 2-regret

152, 25, 98, 151, 122, 3, 109, 46, 72, 166, 92, 8, 172, 189, 124, 4, 49, 33, 47, 140, 154, 161, 26, 125, 170, 115, 113, 155, 141, 60, 157, 45, 134, 175, 62, 91, 36, 197, 132, 103, 178, 65, 145, 28, 149, 13, 37, 116, 42, 50, 130, 195, 121, 198, 96, 48, 164, 43, 32, 89, 44, 83, 57, 79, 0, 135, 162, 119, 104, 77, 12, 194, 190, 70, 199, 165, 169, 105, 61, 193, 196, 10, 148, 131, 118, 110, 23, 93, 9, 14, 20, 74, 17, 171, 99, 163, 88, 82, 167, 85

Greedy cycle, 2-regret + best change

60, 157, 45, 134, 175, 71, 113, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21, 30, 183, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196, 114, 10, 90, 14, 2, 74, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24, 34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 3, 184, 86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 155, 141, 40

Source code

Source code of the compared labs:

- Lab 1: labs/lab1 (Python, C, C++, Rust)
- Lab 2: labs/lab2 (Python, C, C++, Rust)

Conclusions

- Pure 2-regret is much worse (about 122000 on TSPA) because it ignores node costs; adding the best change fixes this.
- 2-regret + best change beats the lab 1 methods: greedy cycle on TSPA (avg 73350), nearest neighbor any position on TSPB (avg 55698).